Example 2.11

The parallel sides of a trapezium are 23 cm and 12 cm. The distance between the parallel sides is 9 cm. Find the area of the trapezium.

Solution

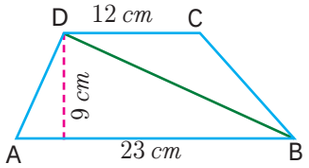


Fig.2.24

Given, height (h) = 9 cm

Parallel sides are (a) = 23 cm and (b) = 12 cm

Area of the trapezium $= \frac{1}{2} \times h\ (a+b)$

$= \frac{1}{2} \times 9\ (23+12)$

$= \frac{1}{2} \times 9\ (35)$

$= 157.5$ $sq.\ cm$

Therefore, Area of the trapezium is 157.5 $sq.\ cm$.

Example 2.12

The area of a trapezium is 828 $sq.\ cm$. If the lengths of its parallel sides are 19.6 cm and 16.4 cm, find the distance between them.

Solution

Given, Area of the Trapezium= 828 cm^2

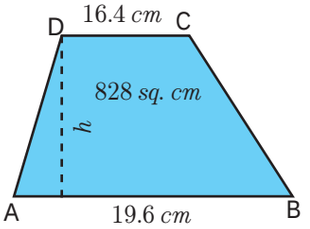


Fig.2.25

$\frac{1}{2} \times h\ (a+b) = 828$

$\frac{1}{2} \times h\ (19.6+16.4) = 828$

$\frac{1}{2} \times h\ (36) = 828$

$h\ (18) = 828$

$h = \frac{828}{18}$

$h = 46\ cm$

Therefore, distance between the parallel sides = 46 cm

Example 2.13

The area of a trapezium is 352 $sq.\ cm$ and the distance between its parallel sides is 16 cm. If one of the parallel sides is of length 25 cm then find the length of the other side.

Solution

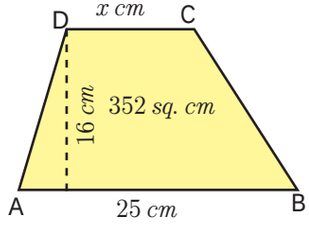


Fig.2.26

Let, the length of the required side be 'x' cm.

Then, area of the trapezium $= \frac{1}{2} \times h\ (a+b)$ $sq.\ units$

$= \frac{1}{2} \times 16\ (25+x)$

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